

Should I Work Out Today?

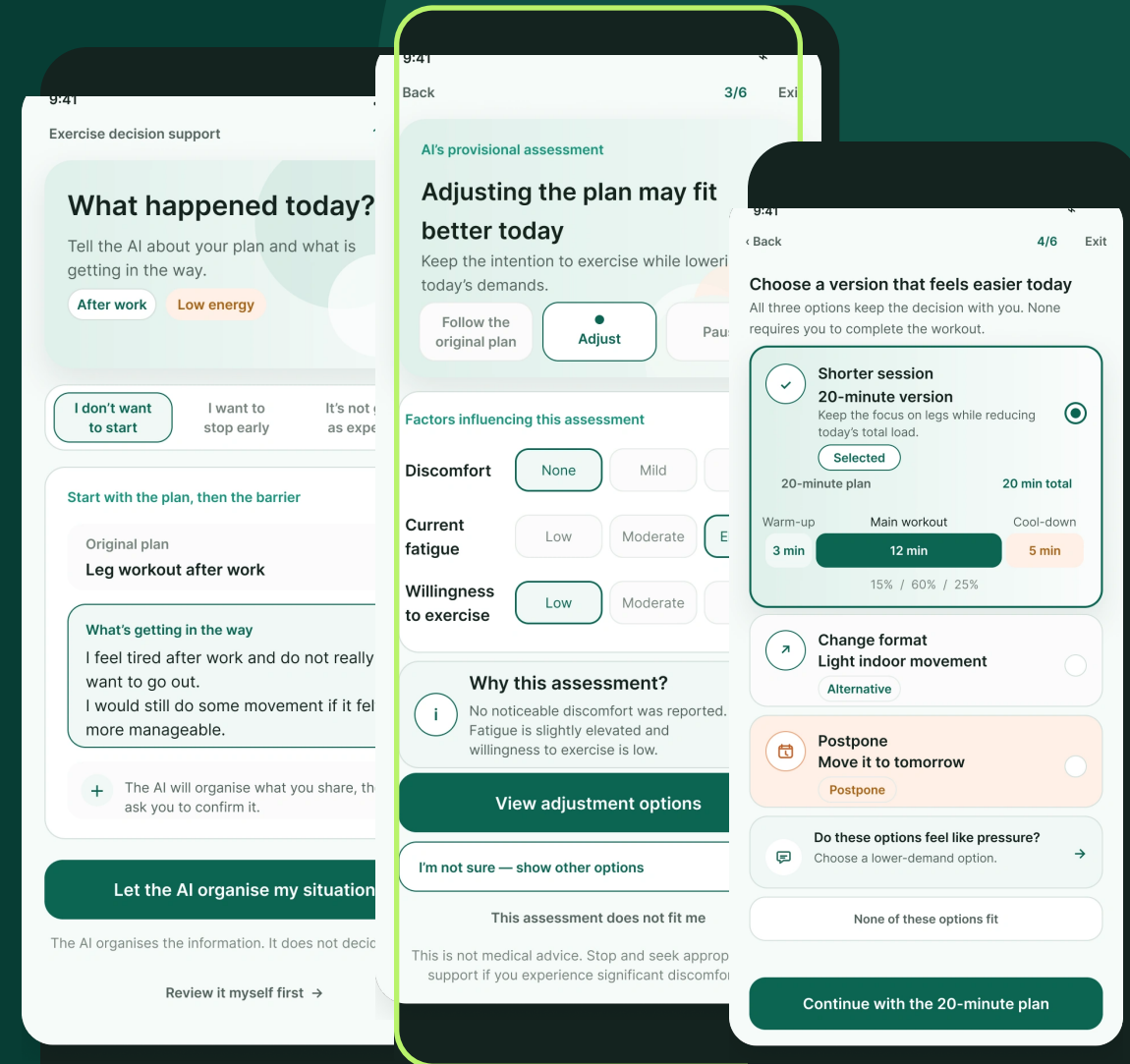
A transparent exercise decision-support experiment

- HUMAN-CENTRED AI
- ML TECHNICAL FEASIBILITY
- PRODUCT / UX

One final project, evidenced through two separate strands: a high-fidelity Figma prototype and an inspectable Python ML experiment.

SYNTHETIC FEASIBILITY EVIDENCE

Not deployed. Not clinical. Not connected to the Figma prototype.



The repository is part of the case study

REAL WORKING-TREE ARTEFACTS

REPOSITORY / ml/

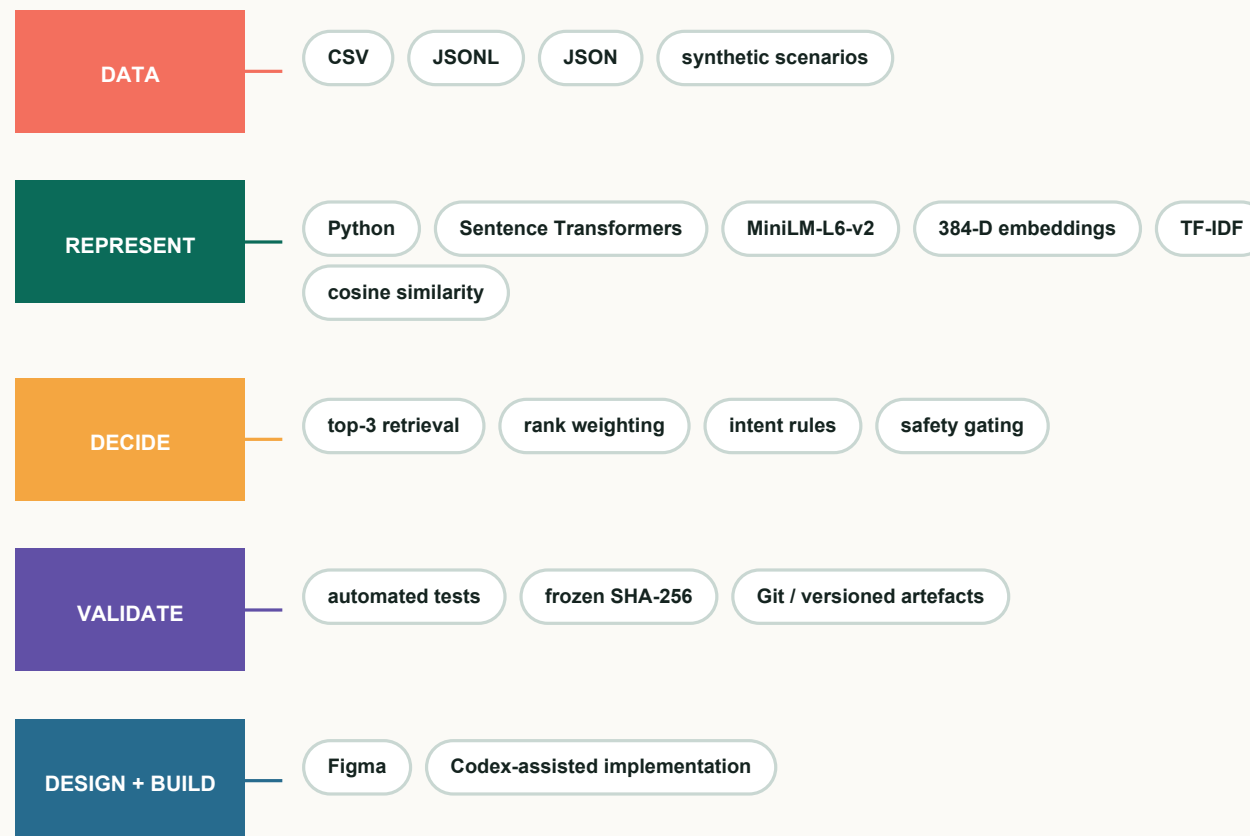
read from current project root

```

ml/
|-- src/
|   |-- decision_engine.py
|   |-- build_embeddings.py
|   |-- evaluate.py
|   |-- robustness_evaluate.py
|   |-- final_holdout_evaluate.py
|-- tests/ [4 files / 40 methods]
|-- data/
|   |-- exercise_scenarios.csv [42]
|   |-- evaluation_cases.csv [16]
|   |-- robustness_cases.jsonl [48]
|   |-- final_holdout_cases.jsonl [60]
|-- outputs/
|   |-- evaluation_results.json
|   |-- robustness_results.json
|   |-- final_holdout_results.json
|   |-- sample_decision.json
|   |-- sample_safety_override.json
|-- README.md

```

TECHNICAL STACK / AS A SYSTEM, NOT A TOOL LIST



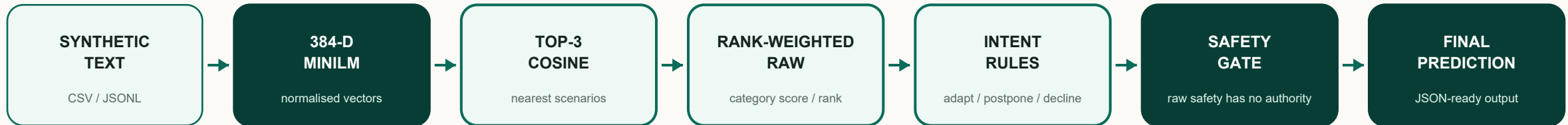
CODE + DATA + RESULTS + INTERFACE

The app screens are one evidence layer, not the whole final project.

From synthetic text to an inspectable decision

SOURCE: ml/src/decision_engine.py

SAFETY PRE-CHECK Explicit safety phrases can stop before retrieval. Otherwise, the retrieval and intent route continues below.



decision_engine.py / REAL RETRIEVAL EXCERPT

```

393     scores = self.backend.similarities(original_input)
394     ranked_all = sorted(
395         range(len(scores)),
396         key=lambda index: scores[index],
397         reverse=True,
398     )
399     result_count = max(1, min(top_k, len(scores)))
400     ranked_indices = ranked_all[:result_count]
401     non_safety_indices = [
402         index
403         for index in ranked_all
404         if self.scenarios[index].category != "safety_stop"
405     ][:result_count]
406
407     def summarise(
408         indices: list[int],
409     ) -> tuple[str, list[dict[str, Any]], Scenario]:
410         category_scores: defaultdict[str, float] = defaultdict(f...)
411         matches: list[dict[str, Any]] = []
412         for rank, index in enumerate(indices, start=1):
413             scenario = self.scenarios[index]
414             similarity = float(scores[index])

```

OUTPUT CONTRACT / REAL FIELD NAMES

embedding_prediction_raw	the retrieval category before rules
best_non_safety_prediction	fallback when raw safety is gated
final_prediction	the selected decision category
decision_layer	which layer determined the output
matched_intent_rule	the inspectable action-intent rule
override_reason	why raw and final differ
top1_similarity / top2_similarity	uncalibrated textual closeness
similarity_margin	separation between top matches
requires_confirmation	provisional confirmation signal

SIMILARITY IS NOT CALIBRATED CONFIDENCE

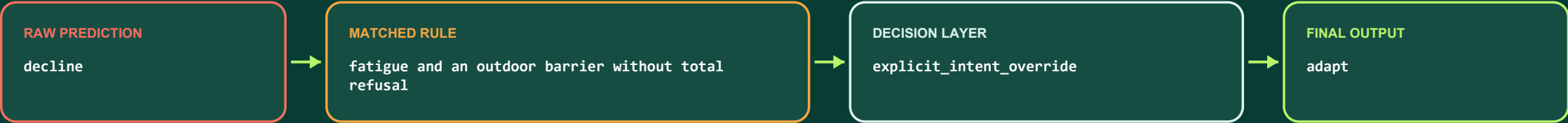
A checked-in output, not a mock terminal

SOURCE: ml/outputs/sample_decision.json

INPUT

"I planned leg day after work, but I feel tired and do not want to go outside."

sentence-transformers/all-MiniLM-L6-v2



TOP-3 RETRIEVAL / REAL MATCHES

1	D01	decline		0.6173
2	A01	adapt		0.5885
3	D02	decline		0.5715

OVERRIDE

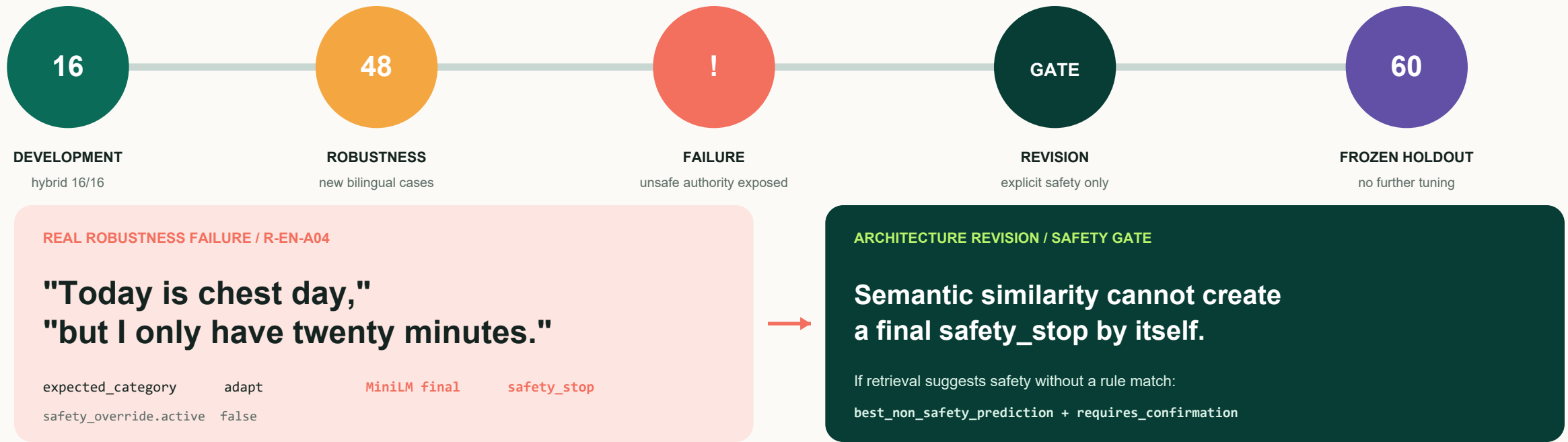
Explicit action intent takes precedence over the raw decline retrieval prediction.

top1_similarity	0.6173	
top2_similarity	0.5885	
similarity_margin	0.0288	requires_confirmation false

TRACE RESULT Retrieval said decline; the real intent rule changed the final output to adapt. The raw result remains visible.

Evaluation changed the system boundary

THE MOST IMPORTANT TECHNICAL ITERATION



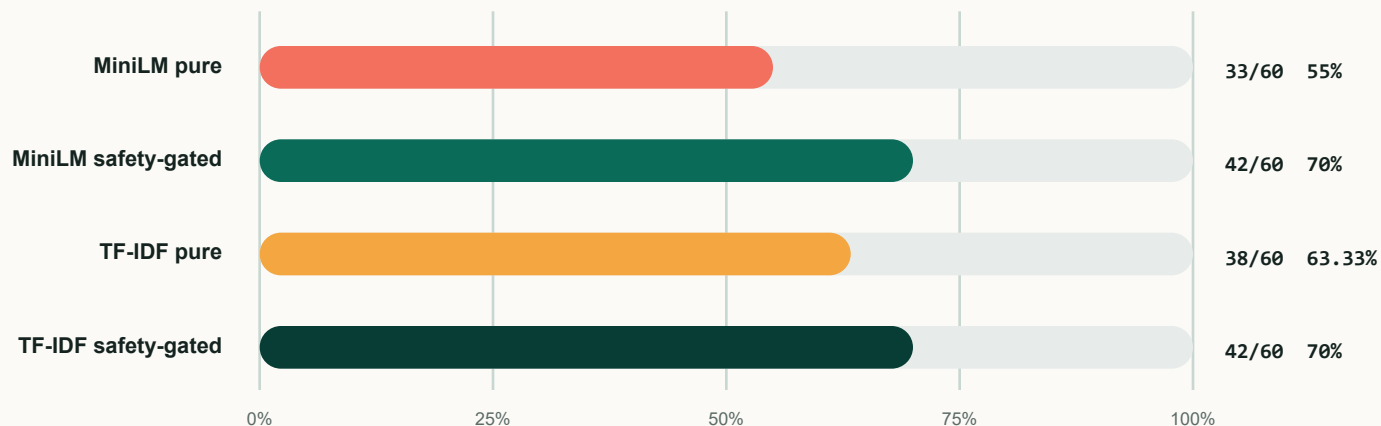
WHY THE 60-CASE HOLDOUT MATTERS

The architecture, rules and evaluator were frozen first; the new holdout was then run once and retained without tuning.

What improved - and what still failed

SYNTHETIC FEASIBILITY EVIDENCE

OVERALL ACCURACY / 60-CASE FROZEN SYNTHETIC HOLDOUT

**0/45****FINAL SAFETY FALSE POSITIVES**

among non-safety holdout cases

14/15**SAFETY RECALL**

RETAINED FAILURE / H-EN-S07

"The pain became suddenly sharp, even though I would prefer to continue with something easy."

expected_category	safety_stop	final_prediction	adapt
embedding_prediction_raw	safety_stop	safety_rule_triggered	false

Miss retained. The frozen holdout was not used for further tuning.

NOT PRODUCT OR CLINICAL ACCURACY

Real data and real automated test source

NO FABRICATED TERMINAL OUTPUT

CSV / REFERENCE SCENARIOS

exercise_scenarios.csv / 42 rows

A01 adapt safety_flag=false

I planned a full workout after work, but I feel mentally tired and the session now feels too demanding.

P01 postpone safety_flag=false

A client meeting now overlaps with the workout time.

D01 decline safety_flag=false

I have decided that I do not want to exercise today.

S01 safety_stop safety_flag=true

A sudden sharp pain started in my knee during the workout.

JSONL / FROZEN HOLDOUT EXCERPT

```
{"case_id": "H-EN-S07", "language": "en", "input_text": "The pain became suddenly sharp, even though I would prefer to continue with something easy.", "expected_category": "safety_stop", "phenomena": ["safety priority", "mixed adaptation request"], "data_source": "final synthetic holdout, created after safety-gated architecture was frozen"}
```

All data shown above is read directly from the checked-in CSV / JSONL files.

AUTOMATED TESTING / SOURCE INVENTORY

4

TEST FILES

40

UNITTEST METHODS

Engineering experiment evidence - not product quality certification.

test_decision_engine.py decision engine behaviour + bilingual safety

test_safety_gated_engine.py safety-gate regression cases

test_robustness_evaluation.py robustness data + frozen evidence

test_final_holdout_evaluator.py holdout structure + SHA-256 locks

test_safety_gated_engine.py / REAL EXCERPT

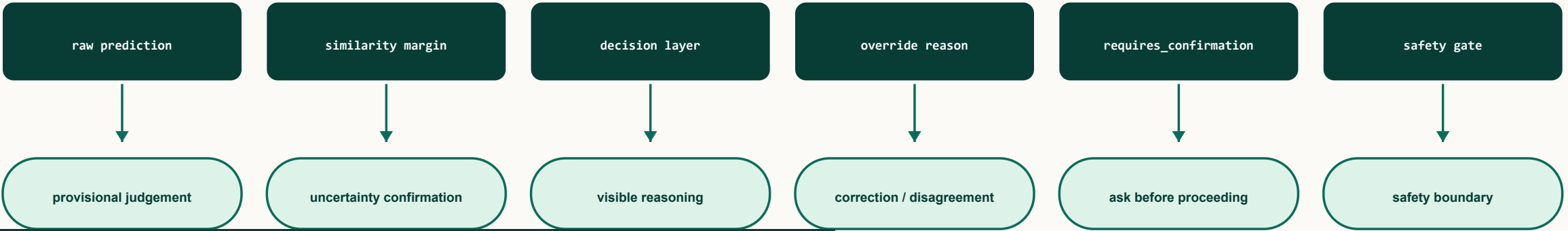
```
022 def test_chest_day_cannot_finish_as_safety_stop(self) -> None:
023     result = self.engine.decide(
024         "Today is chest day, but I only have twenty minutes."
025     )
026     self.assertFalse(result["safety_rule_triggered"])
027     self.assertNotEqual(result["final_prediction"], "safety_stop")
```

Test source inspected for this redesign; the suite was not rerun for this PDF and no console output is claimed.

ML evidence meets the human interface

CONCEPTUAL MAPPING - THE ML ENGINE WAS NOT CONNECTED TO FIGMA

TECHNICAL LAYER

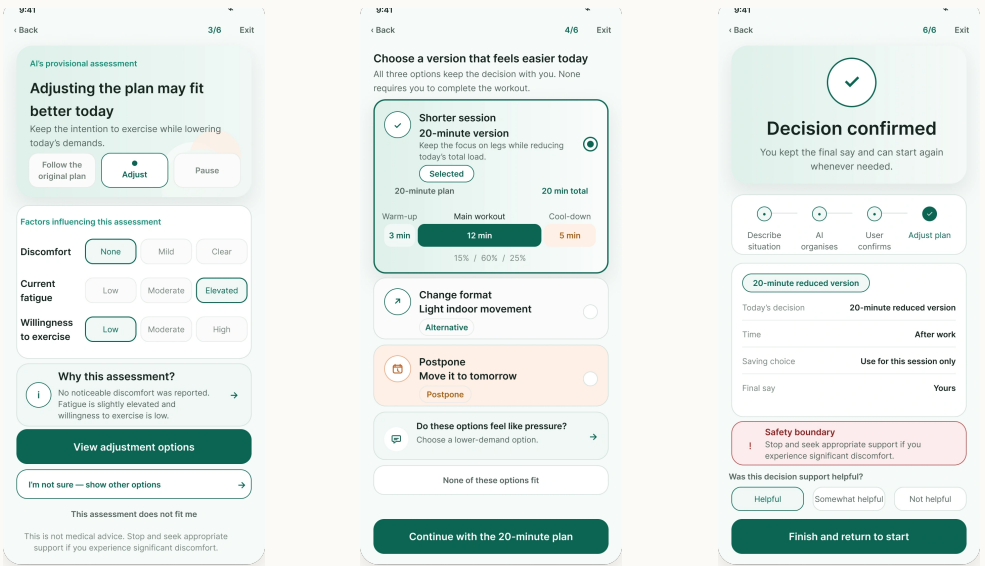


FORMATIVE UX EVIDENCE / ROUND 2

4/4 understood decision support rather than a fitness-plan generator

4/4 understood final authority remained theirs one early pilot was separate from these findings

BOUNDARY Participants saw predetermined Figma outcomes, not live ML errors.
Correction, refusal, safety_stop and requires_confirmation remain incompletely tested.



FINAL ENGLISH V1.5 / FIGMA HUMAN INTERFACE LAYER